Brand
Sight

Coca-Cola vs Pepsi
Brand Visibility Analysis

Factoría F5
AI Bootcamp
2026

YOLOv11 large · Streamlit · Supabase · Gemini AI · OpenCV

AI · Computer Vision · Real-time Analytics

THE PROBLEM

Product Owner

Sponsorships cost millions

Brands invest heavily in product placement without ROI measurement.

Manual review doesn't scale

Frame-by-frame analysis is slow, costly and inherently subjective.

No automated solution exists

There is no tool that measures brand visibility in video content.

The key question

What percentage of screen time does each brand occupy?

THE SOLUTION

BrandSight

Automated brand visibility
analysis powered by AI

- Upload MP4 video — automatic analysis
- YOLOv11 large detects Coca-Cola & Pepsi logos
- Visibility metrics: screen time, %, detections
- Competitive analysis: dominant brand & gap
- Generates AI reports via Gemini

HOW IT WORKS

📹 Upload MP4

↓

🔍 YOLO detects

↓

📊 Metrics

↓

🤖 AI report

↓

💾 Supabase

↓

🌐 Web app

TEAM

Mar Izquierdo
Product Owner

- User stories & README
- Demo videos & documentation
- AI report · Streamlit UI · Slides

Mirae Kang
Scrum Master + Backend

- Agile ceremonies & Kanban
- Git / PRs / deployment
- Pipeline · Supabase · metrics
- Docker · smoke tests

Juan Manuel Iriondo
ML Engineer

- Roboflow dataset & labelling
- Google Colab training
- best.pt · detect_image
- detect_webcam · inference

TECH STACK

Frontend

Streamlit · Python

Computer Vision

YOLOv11 large · OpenCV

Database

Supabase PostgreSQL

Generative AI

Gemini / Jinja2 fallback

Deployment

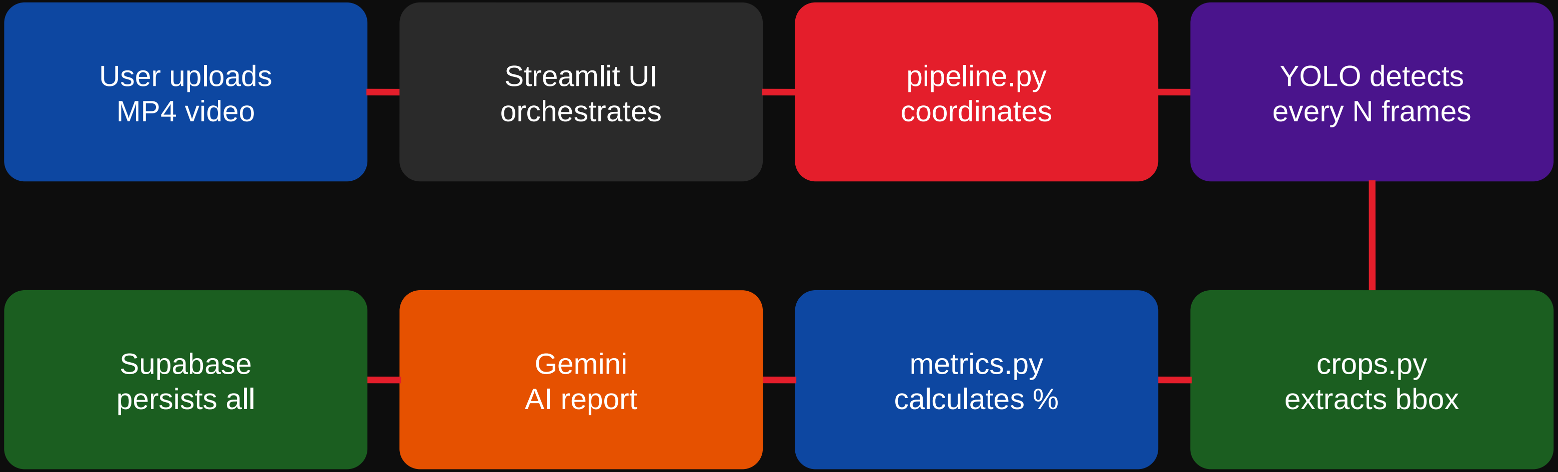
Streamlit Cloud · Docker

Dataset

Roboflow · 2 classes
bbox annotations

Scrum Master

SYSTEM ARCHITECTURE



SAMPLE_STRIDE = 3 · every 3rd frame | **CONFIDENCE_THRESHOLD = 0.5** · min 50% confidence | **5 Supabase tables**

videos **detections** **brand_summary** **competitive_analysis** **marketing_reports**

HOW WE BUILT IT

Agile / Scrum

- ✦ 6-day sprint · 3 devs
- ✦ Daily standup · sprint planning
- ✦ Retrospective at sprint end
- ✦ Kanban on GitHub Projects
- ✦ DoD per delivery level
- ✦ PO guards scope: 2 brands only

Git Workflow

- ✦ main → develop → feature branches
- ✦ One branch per issue
- ✦ Pull Requests + code review
- ✦ SM merges PRs same day
- ✦ smoke_deploy verification
- ✦ Semantic tag: v1.0-demo

LIVE DEMO

ML Engineer

01 **Upload MP4 video (30s)**
Demo selector or custom upload

02 **Frame-by-frame logo detection**
Bounding boxes · brand label · confidence %

03 **Visibility metrics per brand**
Screen time (s) · visibility % · detections

04 **Detection gallery**
Bbox crops saved per detection

05 **AI marketing report**
Gemini · Coca-Cola analyst perspective

🔗

DEMO URL

https://computer-vision-object.streamlit.app/

📹 Webcam segment
detect_webcam.py
local · real-time

MODEL TRAINING

Training Config

Model:	YOLOv11 large (yolo11l.pt)
Dataset:	Roboflow · 2 classes
Classes:	CocaCola · Pepsi
Epochs:	50
Batch / img:	16 · 640 px
Optimizer:	AdamW · lr 0.001
Environment:	Google Colab T4 GPU
mAP50:	**> 0.85 on validation**

Data Augmentation

Horizontal flip	**p = 0.5**
Rotation	**± 15°**
Brightness	**± 25%**
Contrast	**± 25%**
Scale	**± 50%**

Transfer learning from COCO → specialised in 2 classes in 50 epochs

PROCESSING PIPELINE

1 **User uploads MP4 video**

Streamlit uploader · up to 200MB · H.264

2 **YOLO processes every 3 frames**

SAMPLE_STRIDE=3 · CONFIDENCE_THRESHOLD=0.5

3 **Annotated video with bounding boxes**

Brand label + confidence % overlay per frame

4 **Bbox crops extracted**

crops.py · saved to data/crops/{video_id}/

5 **Visibility metrics per brand**

visible_seconds · visibility_pct · avg_confidence

6 **AI report generated**

Gemini API · Jinja2 fallback if API unavailable

7 **Everything saved to Supabase**

5 tables · videos · detections · summaries · analysis · reports

RESULTS & EVALUATION

>0.85	~30s	5	6
mAP50	30s clip	DB tables	scripts
validation accuracy	*processing time (CPU)*	*Supabase PostgreSQL*	*verification & smoke*

All briefing requirements met:

- **Dataset:** Roboflow · 2 classes · bounding box · train/val/test split
- **Preprocessing:** 640×640 resize · automatic normalisation via ultralytics
- **Fine-tuning:** YOLOv11 large pretrained COCO → 50 epochs → 2-class specialist
- **Augmentations:** Flip · Rotation · Brightness · Contrast · Scale · Mosaic
- **Real-time:** detect_webcam.py — live demo segment during presentation

THANK YOU 🙏

GitHub

https://github.com/Bootcamp-IA-P6/Proyecto11_ComputerVision_Equipo1.git

Live demo

https://bit.ly/4oRCO7a

Slides PDF

docs/presentation.pdf

Questions?

Team

Mar Izquierdo — Product Owner
Mirae Kang — Scrum Master
Juan M. Iriondo — ML Engineer

https://bit.ly/4oRCO7a

Developed as an educational project for the AI Development Bootcamp at Factoría F5 · 2026